

Final Project Report: Suicide and Depression Detection Using NLP

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Date: April 30, 2025

1. Introduction

Mental health disorders, particularly suicide and depression, are rising global concerns. Early detection through social platforms like Reddit has become a valuable strategy in identifying at-risk individuals. This project presents a lightweight and interpretable NLP-based classification system that identifies posts containing suicidal ideation using Logistic Regression and TF-IDF. It emphasizes transparency and performance, incorporating LIME for prediction explainability.

2. Problem Definition and Algorithm

2.1 Task Definition

The task is a binary classification problem. Each Reddit post is analyzed and classified as either:

- Suicide (1)
- Non-Suicide (0)

Inputs: Raw Reddit post text

Outputs: Class label indicating suicidal intent

2.2 Algorithm Description

Pipeline:

1. Text cleaning using regex
2. Feature extraction with TF-IDF (10,000 most important terms)
3. Logistic Regression model training
4. Evaluation using metrics (F1-score, AUC)
5. Model interpretability using LIME

Pseudocode:

for each post in dataset:

 clean post

 extract TF-IDF vector

train logistic regression model
predict and evaluate on test set
use LIME to explain predictions

3. Experimental Evaluation

3.1 Methodology

- Dataset: Reddit Suicide Watch Dataset (232,000+ posts)
- Split: 80% training / 20% test
- Vectorization: TF-IDF (stopwords removed)
- Classifier: Logistic Regression
- Metrics: Accuracy, Precision, Recall, F1, AUC

3.2 Results

- Accuracy: 93%
- F1-score: 0.93 for both classes
- AUC: 0.96
- Top Suicide-Related Words: suicide, pills, overdose, rope, die
- Top Non-Suicide Words: crush, teens, minecraft, rant, cute

Confusion Matrix:

	Predicted: Non-Suicide	Predicted: Suicide
Actual: Non-Suicide	21,745	1,463
Actual: Suicide	1,624	21,583

3.3 Discussion

The Logistic Regression model with TF-IDF performed strongly, achieving high recall and precision. Using LIME allowed interpretation of word contributions in predictions, validating that the model's decisions align with psychological patterns in the data. This makes it usable in sensitive, real-world settings.

4. Related Work

Prior work often leverages deep learning (e.g., LSTMs, BERT) for suicide detection. While such models achieve high accuracy, they lack explainability. Our approach offers a balance between performance and interpretability, ideal for healthcare applications where model decisions must be justifiable.

5. Future Work

- Fine-tuning BERT or RoBERTa for multi-class classification
- Adding depression and anxiety as separate classes
- Deploying as a real-time API or integration with Reddit comment monitoring
- Applying emotion classification and unsupervised topic modeling

6. Conclusion

This project developed a fast, accurate, and explainable suicide detection system for Reddit posts. By combining traditional NLP (TF-IDF) with Logistic Regression and LIME, the model achieved over 93% accuracy and demonstrated high interpretability. This makes it ideal for mental health triaging tools.

7. References

<https://www.kaggle.com/datasets/nikhileswarkomati/suicide-watch>